2 Fabrication of MOSFETS

Fabrication Steps for A Silicon Gate NMOS Transistor

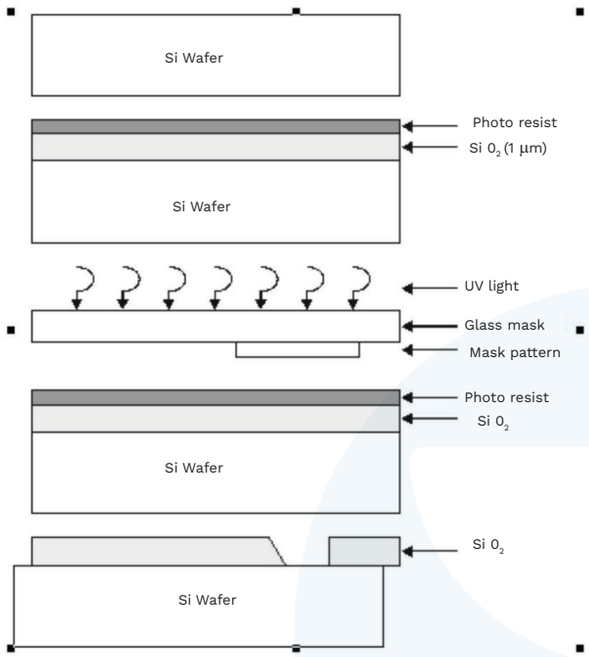
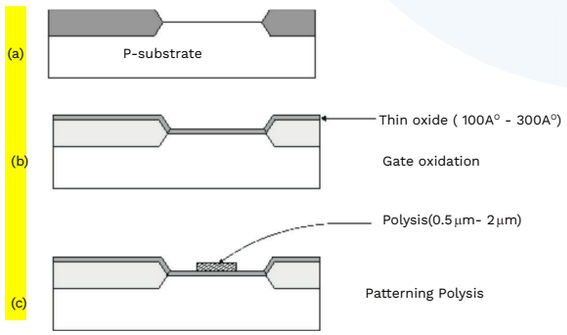


Fig. 2.1 Basic Steps in a Silicon Gate Process

The steps involved in a typical Si gate process entail photomasking and oxide etching as shown in Fig. 2.1.



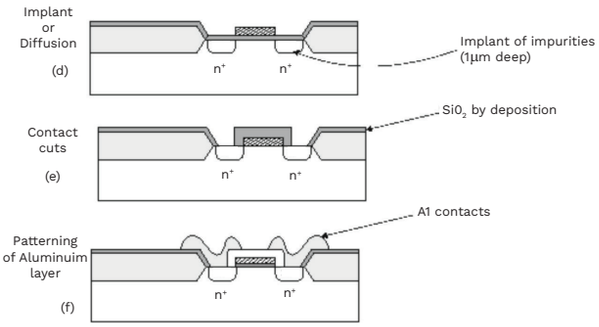


Fig. 2.2 Fabrication Steps for a Silicon Gate nMOS Transistor

- Wafer is covered with SiO_2 of two different thicknesses.
- A thin SiO_2 layer is used where active transistors are required, which is called gate oxide or thinox.
- Thick oxide or field oxide is required to isolate the individual transistors.
- Diffusions are formed only in regions where the polysilicon gate does not shadow the underlying substrate. This is a self-aligned process because the source and the drain do not extend under the gate.
- Finally, the complete structure is covered with SiO_2 and contact holes are etched to make contact with the underlying layers.

Basic CMOS Technologies

- n-well process
- p-well process
- Twin-tube process
- Silicon on insulator

Basic n-well CMOS Process

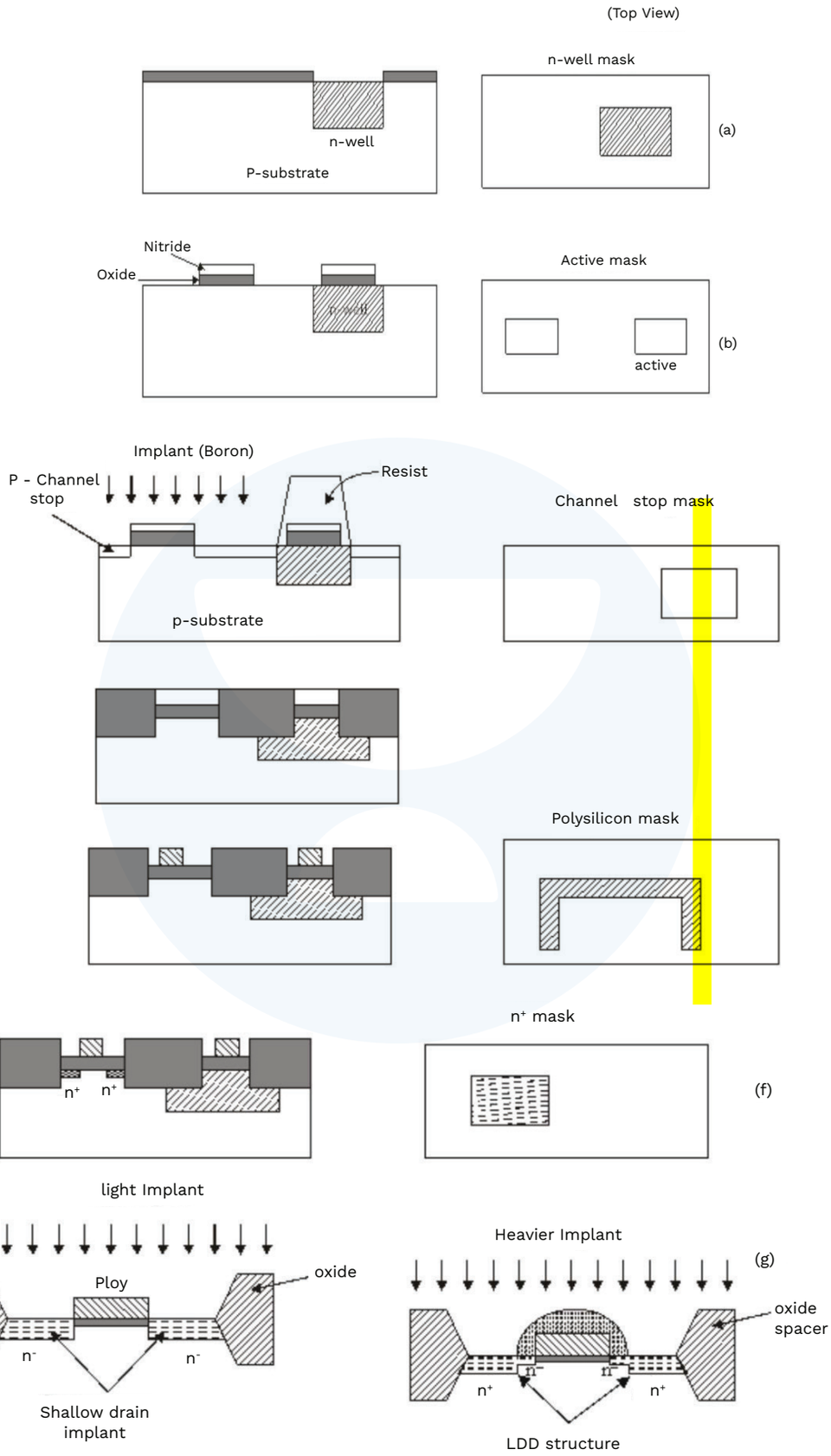


Fig. 2.3 A Typical n-Well CMOS Process

- A shallow n-well is produced using n-well mask. For closely spaced structures, a shallow well is required.
- Active mask defines areas of thin oxide that are required to implement transistor gate and allow implantation to form p or n type of diffusion for transistors a the source or drain region.
    - It is also called thinox mask, thin oxide mask, island, and mesa.
    - SiO_2 is grown and covered with SiN (SiN improves its properties).

Here, SiN acts as a masking layer for the following two steps. Using the channel stop mask, the channel stop implant using boron is done.

- Boron reduces short channel effects in n-channel, MOSFETs.
- Boron reduces the drain electric field penetration, a phenomenon called drain induced barrier lowering (DIBL).
- DIBL is a short channel effect in MOSFETs referring to the reduction of VT of transistor at higher drain voltage.

(i.e., PE barrier for electrons in channel is lowered). (L_{eff} = x_j, L_{eff} = effective channel length, x_j = S/D junction depth).

- This boron implant in conjunction with the thick field oxide that will cover these areas aids in preventing conduction between unrelated transistor 'source' or 'drain'.
- Following the channel-stop implant, thick field oxide is grown where the SiN layer is absent.
    - This oxide grows vertically, as well as, laterally under SiO_2/SiN sandwich. This lateral movement is called the 'bird's beak' because of the shape of oxide encroachment under gate oxide.
    - This general oxide construction technique is called local oxidation of silicon (LOCOS).
    - Techniques, such as side wall masked isolation (SWAMI), reduce the effect of bird's beak.

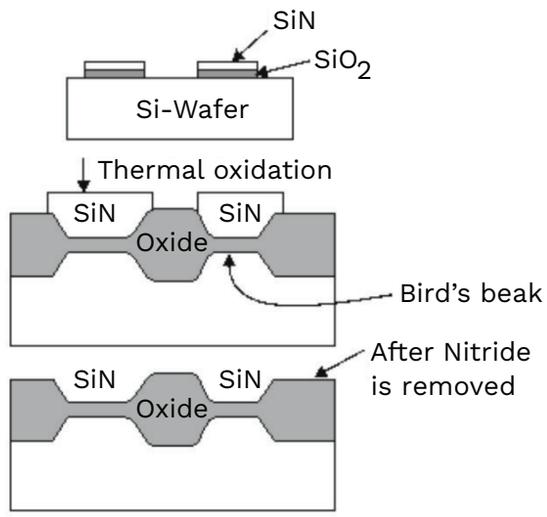


Fig. 2.4 Basic Steps of the LOCOS Process to Create Oxide Isolation Around Active Areas

- Polysilicon is deposited in the required pattern.
- Select mask is used because it selects those transistor regions that are to be n-type.
- In modern small dimension processes, to reduce hot carrier effect, LDD, such as complicated structures are used (LDD is lightly doped drain) as shown in Fig. 2.5(g).
    - Hot carrier effect: charge carriers obtain energy and can become trapped in the gate dielectric of a MOS transistor.
    - The LDD step is not necessarily performed for p-transistors because their hot-carrier susceptibility is much less than that of n-transistors.
- P^+ mask is used to define P^+ diffusions.
- Contact mask is used to create the contact cuts for metallization.
- Metal mask is used for metallization.

p-Well Process

It is preferred when the characteristics of n and p transistors are required to be more balanced than that achievable in an n-well process.

- Since the transistor that resides in the native substrate tends to have better characteristics, p-well process has better p-devices than n-well process.
- Since p-devices have low gain than n-devices, n-well is preferred generally.

Twin-Tube Process

- It consists of both n-well and p-well.
- It provides separate optimisation of threshold voltage, gains body effect, and other characteristics of n and p-devices.

Silicon on Insulator

Advantages over traditional CMOS technology.

- Closer packing of p and n transistors.
- Absence of latch-up problems.
- Lower parasitic substrate capacitances.

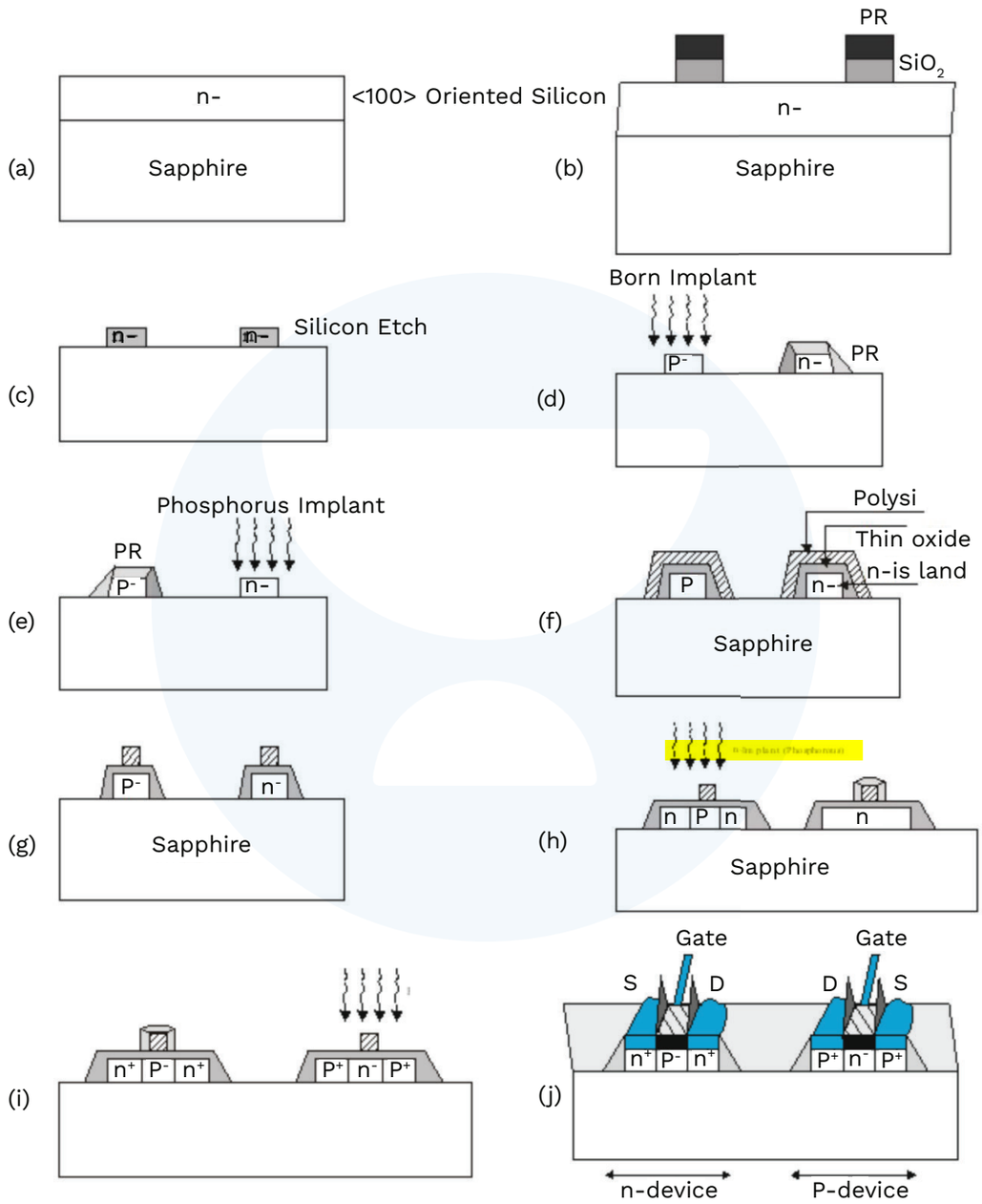


Fig. 2.5 SOI Process Flow

Advantages of SOI Technology

- Due to the absence of wells, transistor structures denser than bulk silicon are feasible.
- Lower substrate capacitances provide the possibility for faster circuits.
- There is no field inversion problem (insulating substrate).
- There is no latch-up because of the isolation of n and p transistors by insulating substrate.
- There is no conducting substrate. Therefore, there is no body effect problem.
- There is enhanced radiation tolerance.

Disadvantages of SOI Technology

- The density advantage is not particularly important because the density is determined by the number and density of metal interconnection layer.
- Expensive and process techniques tend to be less developed than bulk silicon techniques.
- High processing cost and complexity are involved.

Latch-Up and Its Prevention

- Latch-up is defined as the generation of a low impedance path in CMOS chips between the V_{dd} and ground rails due to the interaction of parasitic pnp and npn BJTs.
- These BJTs form a silicon controlled rectifier (SCR) with feedback and virtually short circuit, V_{dd}, to ground, thus, causing excessive current flow and even permanent device damage.

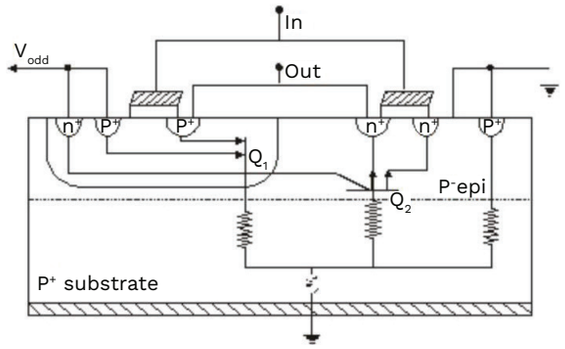


Fig. 2.6 Latch-Up

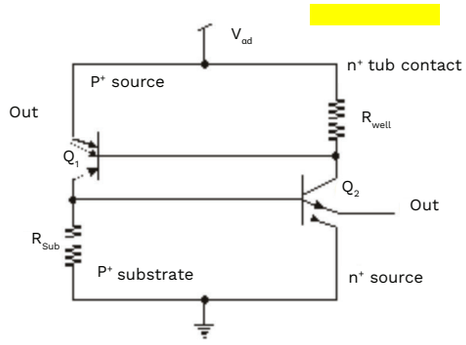


Fig. 2.7 SCR

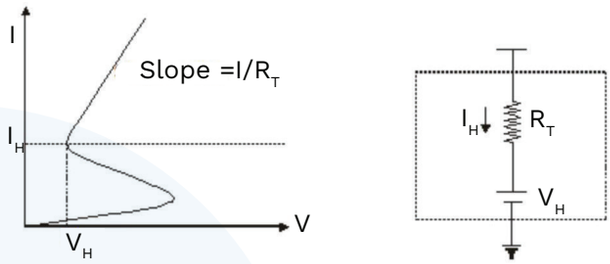


Fig. 2.8 V–I Characteristics of SCR

Causes

- The i/p or o/p signal swings either far beyond the V_{dd} level or far below the V_{SS} level (due to impedance mismatch in transistor lines of high-speed circuits).
- Transistors in V_{dd} or ground buses may turn on a BJT in SCR due to simultaneous switching of many drivers.
- There is leakage of current in well junctions.
- Radiation caused by X-rays, cosmic rays, or particles may generate enough e-h pairs in the substrate and well region, thus, triggering SCR.

Guidelines to Avoid Latch-Up

- Schottky Source/Drain contacts.
- Use P^+ guard rings connected to the ground around nMOS transistors and n+ guard rings connected to V_{dd} around PMOS transistors (to reduce R_W and R_{sub}).
- Place substrate and well contacts as close as possible to the source connections.

- Use minimum area p-wells (in case of twin tub or n-type substrate).
- Use of a lightly doped epitaxial layer on top of a heavily doped substrate shunts lateral currents from the vertical transistor through low resistance substrate.
- NMOS transistors are placed close to V_{ss} and PMOS to V_{dd} rails. Maintain sufficient spacing between PMOS and NMOS transistors.

Practice Exercise

Level I

1. P-well is created on:
(A) n-substrate
(B) n-well
(C) p-substrate
(D) p-well

2. Oxidation process uses:
(A) nitrogen
(B) sulphur
(C) hydrogen
(D) less pure oxygen

3. To isolate individual transistors:
(A) thin oxide is required
(B) thick oxide is required
(C) silicon nitride is required
(D) lithography is required

4. Techniques like SWAMI are used:
(A) for local oxidation of silicon
(B) to reduce short channel effects
(C) to reduce the drain electric field
(D) to reduce the effect of bird's beak

5. Which type of CMOS circuits is generally used?
(A) P-well
(B) n-well
(C) Twin tub
(D) None of the above

6. N^+ and P^+ areas are created in active regions by:
(A) diffusion
(B) ion implantation
(C) CVD
(D) diffusion or ion implantation

7. CMOS technology is used for developing:
(A) digital logic circuits
(B) complex circuits
(C) microprocessors
(D) All of the above

8. CMOS has:
(A) high noise margin
(B) high packing density
(C) high power dissipation
(D) high complexity

9. How many transistors might bring latch-up effect?
(A) Three
(B) Two
(C) Four
(D) One

10. Latch-up can be induced by:
(A) incident radiation
(B) etching
(C) electric field
(D) reflected radiation

Level II

1. Latch-up can be avoided using:
(A) substrate rings
(B) guard rings
(C) latch guard rings
(D) latch-up rings

2. Latch-up problems are highly prone in:
(A) NMOS circuits
(B) PMOS circuits
(C) CMOS circuits
(D) BICMOS circuits

3. Latch-up occurs due to _____ between the power and ground rails.
(A) low impedance path
(B) high impedance path
(C) low resistance path
(D) high resistance path

4. Advantage of Silicon on Insulator (SOI) technology over CMOS technology is:
(A) closer packing of p and n transistors
(B) absence of latch-up problems
(C) lower parasitic substrate capacitances
(D) All of the above

5. DIBL stands for:
(A) Drain Induced Barrier Lowering
(B) Drain Increase Barrier Lowering
(C) Double Increase in Bulk Length
(D) Drain Induced Boron Lowering

6. P-well doping concentration and depth affect the:
(A) drain voltage
(B) V_{DD}
(C) V_{gs}
(D) threshold voltage

7. Lightly doped drain (LDD) structures are used:
(A) in small dimension processes
(B) to reduce hot carrier effect
(C) All of the above
(D) None of the above

8. The gate material generally used is:
(A) silicon
(B) polysilicon
(C) boron
(D) phosphorous

9. Commonly used bulk substrate in nMOS fabrication is:
(A) silicon crystal
(B) phosphorous
(C) boron
(D) gallium

10. Interconnection pattern is made on:
(A) polysilicon layer
(B) silicon dioxide layer
(C) metal layer
(D) diffusion layer

Answer Key

Level I

1. (A) **2.** (C) **3.** (B) **4.** (D) **5.** (B) **6.** (D) **7.** (D) **8.** (B) **9.** (B) **10.** (A)

Level II

1. (B) **2.** (C) **3.** (A) **4.** (D) **5.** (A) **6.** (D) **7.** (C) **8.** (B) **9.** (A) **10.** (C)